

5. Instrumentation

└ 5.2 Alerting and metrics

└ 5.3.1 Inspect: CPU, memory, disk, network

-
1. What tools are available in Azure to inspect CPU, memory, disk, and network metrics?
 2. How do you enable and configure monitoring for infrastructure metrics in Azure?
 3. Where are infrastructure metrics (CPU, memory, disk, network) visualized in the Azure portal?
 4. How do you set up alerts for CPU, memory, disk, or network thresholds in Azure Monitor?
 5. What is the difference between Azure Metrics and Azure Logs for infrastructure monitoring?
 6. How can Application Insights be used for infrastructure telemetry?
 7. What are best practices for monitoring high CPU usage in Azure VMs?
 8. How do you collect and analyze performance counters for Azure resources?
 9. What are the recommended retention and sampling settings for metrics?
 10. How do you diagnose and troubleshoot resource bottlenecks using collected telemetry?
-

1. What tools are available in Azure to inspect CPU, memory, disk, and network metrics?

Azure Monitor and *Azure Log Analytics* provide core infrastructure metrics. *VM Insights* and *Container Insights* give detailed metrics for VMs and containers.

2. How do you enable and configure monitoring for infrastructure metrics in Azure?

Enable monitoring by installing the *Azure Monitor Agent* or *Log Analytics* agent on your resources. Register your VM or container to a *Log Analytics* workspace and enable Insights as required.

3. Where are infrastructure metrics (CPU, memory, disk, network) visualized in the Azure portal?

Metrics are visualized in *Azure Monitor* under “Metrics” or via “Insights” tabs on VM/Container resources. Dashboards can aggregate multiple metrics for quick visibility.

4. How do you set up alerts for CPU, memory, disk, or network thresholds in Azure Monitor?

Create an Alert Rule in *Azure Monitor*. Define a condition (e.g., CPU > 80%), set the evaluation period, and assign an action group for notifications (email, SMS, webhook, etc.).

5. What is the difference between Azure Metrics and Azure Logs for infrastructure monitoring?

Metrics provide near real-time numeric time series (e.g., CPU percentage), optimized for performance. Logs provide richer, queryable telemetry (e.g., event details), useful for diagnostics and historical analysis.

6. How can Application Insights be used for infrastructure telemetry?

Application Insights primarily collects app-level telemetry, but can be extended to include infrastructure metrics by connecting to *Azure Monitor* or configuring custom telemetry ingestion.

7. What are best practices for monitoring high CPU usage in Azure VMs?

Set alerts for sustained high CPU usage, review resource utilization in *Metrics Explorer*, investigate running processes, and scale up/out as required.

8. How do you collect and analyze performance counters for Azure resources?

Enable collection of performance counters using the Log Analytics agent. Use *Log Analytics* workspace to query and analyze the data (e.g., Perf table for Windows/Linux metrics).

9. What are the recommended retention and sampling settings for metrics?

Use default retention (typically 93 days for metrics, 30 days for logs, extendable as needed). Set sampling for logs and custom metrics to balance cost and granularity.

10. How do you diagnose and troubleshoot resource bottlenecks using collected telemetry?

- Correlate spikes in CPU, memory, disk, or network metrics with application and OS logs.
- Use Log Analytics queries to identify patterns and root causes.
- Review recommendations in *Azure Advisor* and *Insights* for optimization actions.